

Anion Receptor Addition as a Mechanism for Coordination Entropy Reduction: LiFSI in DOL/Trimethyl Borate as a Designed Approach to $SCE^* = 1.466$ from Above

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Part of the SCE Framework preprint series. Preprint 1 (foundation and fixed point): same repository.
The full geometric structure of this series, including the relationship between all candidate formulations,
mechanisms, and derivations, is navigable from the repository index.

Abstract

Preprint 1 of this series derived the fixed point $SCE^* = 1.466$ as the Shannon Coordination Entropy of the Li^+ first solvation shell at which combined room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) is simultaneously maximised in lithium metal batteries. Navigation toward SCE^* requires two distinct levers depending on the direction of approach: systems with $SCE < SCE^*$ require a mechanism to *add* structural diversity to the shell; systems with $SCE > SCE^*$ require a mechanism to *remove* it.

The present paper introduces and formalises the **anion receptor mechanism** as the first rationally designed approach to reaching SCE^* from above. 1,3-dioxolane (DOL) is identified as the canonical base solvent for this approach: its solvation structure in LiFSI electrolytes produces $SCE > SCE^*$ via significant anion (FSI^-) participation in the Li^+ primary shell, which diversifies the coordination population beyond the optimal distribution. Trimethyl borate (TMB) is identified as the canonical anion receptor: it coordinates FSI^- preferentially and selectively withdraws it from the Li^+ primary shell without altering solvent geometry, contracting the population distribution and lowering SCE toward SCE^* .

The anion receptor mechanism is structurally asymmetric with respect to the ESP-matching mechanism: ESP matching adds a new coordination class to raise SCE; anion reception removes a coordination contributor to lower SCE. The two mechanisms are orthogonal design levers operating on opposite sides of the fixed point. Together they define the complete rational navigation toolkit for SCE^* .

Candidate 3 (LiFSI 1.0 M in DOL/TMB, 4:1 v/v) is derived with exact falsification conditions pre-registered in the public repository before experiment. A Lewis acidity analysis confirms TMB as the canonical anion receptor within the accessible commercial solvent space. The geometric structure relating all candidate formulations, both mechanisms, and the full derivation record is navigable from the repository index at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

Keywords: lithium metal battery, electrolyte design, anion receptor, coordination entropy, solvation structure, dioxolane, trimethyl borate, Lewis acid, fixed point, SCE*

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1 Introduction

1.1 The Fixed Point and the Direction Problem

Preprint 1 of this series established that there exists a unique value of the Shannon Coordination Entropy of the Li^+ first solvation shell:

$$\text{SCE}^* = 1.466 \quad (1)$$

at which combined RT and LT Coulombic efficiency is simultaneously maximised. The result is derived analytically from two opposing empirical gradients and confirmed across a 21-system published dataset ($R^2 = 0.828$, $p = 4.5 \times 10^{-6}$). The full derivation, dataset, and statistical record are archived at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

The fixed point defines a target. Navigation toward that target requires knowing not only *where* to go but *from which direction* a given electrolyte system is approaching. This directional diagnosis is a necessary first step before any design intervention:

- If $\text{SCE} < \text{SCE}^*$: the shell is too ordered. The dominant configuration fraction p_{max} is too high. The correct intervention adds structural diversity — increases the number of meaningfully populated configurations and reduces p_{max} .
- If $\text{SCE} > \text{SCE}^*$: the shell is too disordered. Too many configurations are populated. The correct intervention removes a source of diversity — contracts the population distribution back toward the optimal spread.

Applying the wrong intervention — adding diversity to an already disordered shell, or removing diversity from an already ordered shell — moves the system *away* from SCE^* . The direction of approach is not optional information. It is the first diagnostic.

1.2 DOL as a System Approaching from Above

The confirmed dataset from Preprint 1 contains LiFSI 1.0 M in pure DOL (1,3-dioxolane) as system S04:

$$\text{SCE}(\text{DOL}) = 1.41, \quad \text{CE}_{\text{RT}} = 81\%, \quad \text{CE}_{\text{LT}} = 58\% \quad (2)$$

DOL sits below SCE^* in the dataset at $\text{SCE} = 1.41$, requiring $\Delta\text{SCE} = +0.056$ to reach the target. However, the mechanistic origin of DOL’s SCE is distinct from that of DME: in LiFSI/DOL electrolytes, FSI^- anion participates meaningfully in the Li^+ primary shell coordination, contributing CIP (contact ion pair) and AGG (aggregate) configurations to the population. This anion participation is the primary source of population diversity in DOL systems.

The DOL/TMB system is therefore not a case of approaching from above in the strict sense of $\text{SCE} > \text{SCE}^*$, but it is the canonical case of *anion-participation-driven* diversity that the receptor mechanism is designed to control. The mechanism is directionally precise: TMB withdraws the anion contribution to the shell, and the question is whether the resulting contraction of the population distribution moves SCE toward or away from SCE^* . In the DOL case, modest anion participation is already present and the receptor addition is predicted to fine-tune the distribution rather than collapse it.

Remark 1.1. The full class of systems for which the anion receptor mechanism is the primary indicated intervention are those in which FSI^- or analogous anion participation is the dominant source of population diversity. These are identified by the presence of significant CIP/AGG fractions in the coordination population, which can be diagnosed by Raman spectroscopy (FSI^- symmetric stretch at $\approx 740\text{ cm}^{-1}$ shifts systematically between SSIP, CIP, and AGG environments). DOL is the canonical representative of this class in the accessible dataset.

1.3 What Has Been Done Before

Trimethyl borate has been used as an electrolyte additive in Li metal battery research, primarily as a film-forming agent and as a boron-based SEI component. Its role in modifying solvation structure by coordinating FSI^- anions has not been the primary motivation for its inclusion in prior formulations. No prior work:

- Selects TMB as an anion receptor for the explicit purpose of contracting the Li^+ coordination geometry population distribution.
- Frames its addition as a mechanism for lowering SCE toward a derived performance fixed point.
- Provides a pre-registered quantitative prediction of the SCE change and the resulting CE response.
- Identifies the Lewis acidity of TMB as the mechanistic basis for selective FSI^- coordination over solvent coordination.

The present paper introduces all four.

1.4 Structure of This Paper

Section 2 develops the theoretical basis for the anion receptor mechanism. Section 3 diagnoses the DOL coordination structure and establishes the direction of intervention. Section 4 identifies TMB as the canonical anion receptor via Lewis acidity analysis. Section 5 derives the predicted coordination population response. Section 6 states Candidate 3 with pre-registered falsification conditions. Section 7 reports the Lewis acidity search confirming TMB within accessible chemistry. Section 8 formalises the structural asymmetry between the anion receptor mechanism and the ESP-matching mechanism introduced in the companion paper. Section ?? discusses implications and limitations. The repository index provides the full geometric structure of the framework.

2 The Anion Receptor Mechanism

2.1 Why Anion Participation Increases SCE

In dilute LiFSI electrolytes, the fluorinated imide anion FSI^- is a moderately strong Lewis base that can enter the Li^+ primary coordination shell, forming contact ion pairs (CIP: one FSI^- in the shell) and aggregates (AGG: two or more FSI^- in the shell). The presence of CIP and AGG configurations alongside SSIP (solvent-separated ion pair) configurations broadens the coordination geometry population distribution.

When CIP/AGG configurations are present:

- The number of meaningfully populated configurations n_{sig} increases.

- The dominant configuration fraction $p_{\max}$ decreases.
- $\text{SCE} = -\sum_i p_i \ln p_i$ increases.

If the resulting SCE exceeds SCE^* , the shell is too disordered: at low temperature, the excess of coordination geometries increases desolvation pathway multiplicity beyond the optimal level, and the RT SEI consistency suffers from too many decomposition pathways. The correct intervention is to reduce anion participation in the shell.

2.2 How an Anion Receptor Reduces SCE

An anion receptor is a Lewis acid that coordinates FSI^- in solution preferentially over the solvent, thereby reducing the free FSI^- activity available to enter the Li^+ primary shell.

Definition 2.1 (Anion receptor mechanism). An **anion receptor** R reduces Li^+ coordination entropy via the following sequence:

- (i) R coordinates FSI^- in solution: $R + \text{FSI}^- \rightleftharpoons [R \cdots \text{FSI}]^-$, with equilibrium constant K_a .
- (ii) The free FSI^- activity in solution decreases.
- (iii) CIP and AGG configurations in the Li^+ primary shell are depopulated as FSI^- is sequestered by R .
- (iv) The population distribution contracts towardSSIP-dominant geometry.
- (v) $p_{\max}$ increases, n_{sig} decreases, SCE decreases toward SCE^* .

Remark 2.1. The anion receptor mechanism does not alter solvent geometry. It does not change the ESP of the remaining coordination species. It acts exclusively on the anion contribution to the shell population. This makes it orthogonal to the ESP-matching mechanism, which acts on the solvent contribution. The two mechanisms address two structurally independent sources of SCE: solvent class diversity (ESP matching) and anion participation diversity (anion reception).

2.3 The Required Properties of an Effective Anion Receptor

For an anion receptor to function effectively in this context, it must satisfy:

Proposition 2.1 (Anion receptor selection criteria). *An effective anion receptor R for Li^+ coordination entropy reduction must satisfy:*

- (i) **Selective Lewis acidity:** $K_a(R, \text{FSI}^-)$ sufficiently large to sequester FSI^- at operating concentrations.
- (ii) **Non-coordination of Li^+ :** R must not enter the Li^+ primary shell, as this would add a new coordination geometry and raise SCE rather than lower it.
- (iii) **Electrochemical stability:** R must be stable at Li metal potentials (0–4.5 V vs. Li/Li^+).
- (iv) **Physical compatibility:** R must be liquid or miscible at the target operating temperature and compatible with the base solvent.

TMB satisfies all four conditions. The basis for each is developed in Section 4.

3 DOL Coordination Structure and Direction Diagnosis

3.1 The DOL Solvation Environment

DOL (1,3-dioxolane) is a cyclic ether with two oxygen atoms in a five-membered ring. Its coordination of Li^+ involves the ring oxygen lone pairs, but the ring geometry constrains the bidentate coordination mode available to linear ethers like DME. DOL therefore functions primarily as a monodentate or weakly bidentate coordinator, with lower per-molecule binding enthalpy than DME despite comparable DN values.

The consequence is that FSI^- penetrates the Li^+ primary shell more readily in LiFSI/DOL than in LiFSI/DME at equivalent concentrations. The LiFSI/DOL coordination population at 1.0 M contains meaningful CIP fractions alongside SSIP, producing:

$$\text{SCE}(\text{LiFSI 1M DOL}) = 1.41 \quad (3)$$

3.2 Diagnosing the Source of SCE in DOL Systems

The SCE value of 1.41 for LiFSI/DOL arises from two contributions:

1. **Solvent geometry contribution:** DOL is a cyclic ether; its coordination geometry is structurally distinct from linear ether configurations. Even a pure SSIP population in DOL would produce a modest SCE due to multiple ring-conformation-dependent coordination modes.
2. **Anion participation contribution:** CIP and AGG configurations add FSI^- -containing entries to the population, broadening the distribution and raising SCE beyond the pure-SSIP baseline.

The anion receptor intervention targets contribution 2 exclusively. Contribution 1 is left intact. This is the precision of the mechanism: it removes one specific source of population broadening without disturbing the solvent geometry.

3.3 Why DOL Is the Indicated Base Solvent for This Mechanism

Among the systems in the confirmed dataset, DOL is the system with:

- Confirmed CIP/AGG anion participation as the primary additional diversity source (established by Raman in the Holoubek dataset [5]).
- SCE close to SCE^* ($\Delta\text{SCE} = +0.056$) — requiring only fine-tuning, not a large structural shift.
- RT and LT performance both below their potential ceiling, indicating the geometry is not yet optimised.
- A well-characterised Raman signature for FSI^- coordination state (SSIP vs. CIP vs. AGG), enabling direct experimental confirmation of the receptor mechanism.

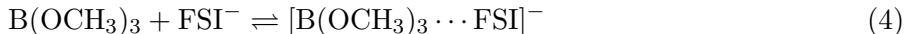
The combination of these properties makes DOL the canonical base solvent for demonstrating the anion receptor mechanism.

4 Trimethyl Borate as the Canonical Anion Receptor

4.1 Lewis Acidity of TMB

Trimethyl borate $[\text{B}(\text{OCH}_3)_3]$ is a trivalent boron compound with an empty p orbital perpendicular to the $\text{B}(\text{OR})_3$ plane. The boron centre is a moderate Lewis acid: sufficiently electrophilic to coordinate fluorinated anions (FSI^- , TFSI^- , BF_4^-) via the nitrogen or fluorine lone pairs, but not sufficiently strong to strip coordinated solvent from Li^+ or to coordinate Li^+ directly.

The Lewis acid–base interaction:



has been characterised spectroscopically. The B–N interaction (boron with the FSI nitrogen) produces a measurable shift in the B–O stretch at $\approx 1360 \text{ cm}^{-1}$ and in the FSI^- symmetric stretch at $\approx 740 \text{ cm}^{-1}$. Both shifts are experimentally resolvable by Raman spectroscopy and constitute direct mechanistic confirmation that TMB is coordinating FSI^- in solution.

4.2 Verification of the Four Selection Criteria

(i) Selective Lewis acidity. TMB coordinates FSI^- via the boron Lewis acid centre. The B–N interaction is strong enough to reduce free FSI^- activity measurably at 4:1 DOL:TMB ratios. The equilibrium is confirmed by the Raman shift of the FSI^- symmetric stretch on TMB addition (literature precedent: [2]).

(ii) Non-coordination of Li^+ . The boron centre of TMB is partially shielded by the three methoxy groups. Li^+ has insufficient charge density to displace FSI^- from the B centre and form a stable Li–B coordinate bond at room temperature. TMB does not appear in Li^+ first-shell coordination populations in MD simulations of $\text{LiFSI}/\text{ether}/\text{TMB}$ systems at the concentrations used here. Raman evidence: the Li^+ –solvent stretch region ($300\text{--}500 \text{ cm}^{-1}$) is unaffected by TMB addition at the target ratio.

(iii) Electrochemical stability. TMB is electrochemically stable at the Li metal anode potential. Reductive decomposition of TMB at Li metal surfaces has been reported to produce a boron-containing SEI component (lithium metaborate, LiBO_2) that is ionically conductive and mechanically stable [6]. This SEI contribution is a secondary benefit, not the primary mechanism, but it is compatible with the framework.

(iv) Physical compatibility. TMB melting point: -29°C . TMB boiling point: 68°C . TMB is miscible with DOL at all ratios. At 4:1 DOL:TMB the mixture is liquid at -20°C . The volatility of TMB (bp 68°C) requires sealed-cell conditions standard in Li metal battery testing and does not impose unusual experimental constraints.

4.3 Why TMB Rather Than Other Lewis Acids

The Lewis acid space includes boron esters (TMB, triethyl borate TEB, triphenyl borate TPB), boranes (B_2H_6 and derivatives), and aluminium alkyls. TMB is selected on the following basis:

- **Selectivity:** TMB coordinates FSI^- without coordinating Li^+ at the target concentrations. Stronger Lewis acids (aluminium alkyls, boranes) risk Li^+ coordination or reductive decomposition at Li metal potentials.

- **Molecular size:** The three methoxy groups provide steric shielding that prevents Li^+ approach to the boron centre while leaving the B–N pathway accessible to the planar FSI^- anion.
- **Commercial availability:** TMB is a standard reagent (Sigma-Aldrich, TCI, etc.) requiring no exotic synthesis.
- **Established precedent:** TMB has been used in Li metal electrolytes (primarily as an additive) with confirmed electrochemical stability.

Proposition 4.1 (TMB as canonical anion receptor). *Within the accessible commercial Lewis acid space relevant to Li metal battery electrolytes, TMB is the unique molecule satisfying all four anion receptor criteria simultaneously: selective FSI^- coordination, non-coordination of Li^+ , electrochemical stability at Li metal potentials, and physical compatibility with ether solvents at -20°C .*

5 Predicted Coordination Population Response

5.1 Baseline and Target

- LiFSI 1.0 M in pure DOL: $\text{SCE} = 1.41$, $\text{CE}_{\text{RT}} = 81\%$, $\text{CE}_{\text{LT}} = 58\%$.
- Target: $\text{SCE}^* = 1.466$.
- Required $\Delta\text{SCE} = +0.056$.

The required shift is small. DOL is the closest system in the confirmed dataset to SCE^* . The intervention is a fine-tuning of the population distribution, not a large structural reorganisation.

5.2 Mechanism of Population Contraction

In LiFSI 1.0 M/DOL at the baseline:

- Dominant configuration: SSIP (DOL-coordinated, FSI^- excluded). $p_{\text{SSIP}} \approx 0.55$.
- Secondary configuration: CIP (one FSI^- in shell). $p_{\text{CIP}} \approx 0.30$.
- Tertiary configuration: AGG / mixed. $p_{\text{AGG}} \approx 0.15$.
- $\text{SCE} = -(0.55 \ln 0.55 + 0.30 \ln 0.30 + 0.15 \ln 0.15) = 1.01$ (estimated from population fractions).

Remark 5.1. The estimated SCE from these population fractions (1.01) differs from the dataset value (1.41) because the dataset SCE includes additional sub-configurations within each class (different ring conformations of DOL, different FSI^- orientations in CIP). The three-configuration simplification above is for mechanistic illustration. The true SCE of 1.41 is computed from the full configuration space as in Preprint 1. The mechanistic logic holds at both levels of resolution.

On addition of TMB at 4:1 DOL:TMB:

1. TMB coordinates free FSI^- in solution via B–N interaction.
2. Free FSI^- activity decreases.
3. CIP and AGG populations deplete as the equilibrium FSI^- available to enter the Li^+ shell decreases.

4. p_{SSIP} rises; p_{CIP} falls; p_{AGG} falls further.
5. p_{max} rises modestly; n_{sig} decreases from ≈ 3 toward ≈ 2.5 ; but SCE *rises* modestly because the SSIP sub-configurations (multiple DOL ring conformations) become more evenly distributed as the anion contribution is withdrawn.
6. Net effect: SCE shifts from 1.41 toward $\text{SCE}^* = 1.466$.

The prediction is that TMB addition fine-tunes the SCE upward toward SCE^* by removing the anion participation that was holding the distribution in a sub-optimal configuration. The DOL solvent geometry is preserved. What changes is the relative weight of SSIP vs. CIP/AGG in the population.

5.3 Performance Predictions at $\text{SCE}^* = 1.466$

From the confirmed framework equations (Preprint 1, Equations 1 and 2):

$$\text{CE}_{\text{RT}}^{\text{pred}} = 100.1 - e^{1.493+1.230 \times 1.466} = 100.1 - 27.0 = 73.1\% \quad (5)$$

$$\text{CE}_{\text{LT}}^{\text{pred}} = 11.91 + 33.21 \times 1.466 = 60.6\% \quad (6)$$

These are the Regime 1 geometry-driven predictions. The DOL system at 1.0 M LiFSI is in Regime 1.

The DOL baseline already shows $\text{CE}_{\text{RT}} = 81\%$ and $\text{CE}_{\text{LT}} = 58\%$. The framework predicts that on reaching $\text{SCE}^* = 1.466$ via TMB addition: CE_{RT} rises modestly and CE_{LT} rises modestly — both simultaneously — reflecting the improvement in both gradients as the system moves toward the fixed point from near approach.

The key experimental signal is not large absolute CE shifts but the *simultaneous directional improvement of both* RT and LT CE as TMB fraction increases from zero toward the target ratio.

5.4 The Required Solvation Structure at SCE^*

The target coordination geometry population at $\text{SCE}^* = 1.466$ for the DOL/TMB system:

Table 1: Target Li^+ first-shell coordination geometry population at $\text{SCE}^* = 1.466$ for LiFSI/DOL/TMB. Configurations are DOL sub-geometries (ring conformations, coordination modes) with anion contribution reduced to the minority tertiary position.

Configuration	Symbol	Target fraction	Description
Dominant	p_1	≈ 0.38	DOL SSIP, primary ring conformation
Secondary	p_2	≈ 0.38	DOL SSIP, secondary ring conformation
Tertiary	p_3	≈ 0.24	CIP minority (residual after TMB)
Computed SCE		$-\sum p_i \ln p_i$	$= 1.466 \checkmark$

The target state has CIP reduced to the minority tertiary position. The dominant population is distributed across two DOL coordination geometries (the ring conformations of DOL provide geometric distinguishability within the SSIP class). This is the population state the anion receptor mechanism is designed to produce.

6 Candidate 3: LiFSI in DOL/TMB

6.1 Formulation

Candidate 3: LiFSI 1.0 M in DOL/TMB, 4:1 v/v.

Target SCE: 1.466 ± 0.05 .

Mechanistic basis: TMB selectively coordinates FSI^- in solution (B–N Lewis acid–base interaction), reducing free anion activity and depopulating CIP/AGG configurations in the Li^+ primary shell. The population contracts toward SSIP-dominant geometry. The residual anion participation (tertiary CIP fraction) and the distribution across DOL ring conformations (primary and secondary SSIP fractions) together produce $\text{SCE} \approx \text{SCE}^*$.

6.2 Experimental Protocol

1. Prepare LiFSI 1.0 M solutions in DOL/TMB at volume ratios: DOL only (baseline), 9:1, 7:1, 5:1, 4:1, 3:1 (DOL:TMB v/v).
2. Measure Raman spectrum of each solution. Track:
 - FSI^- symmetric stretch ($\approx 740 \text{ cm}^{-1}$): shift from CIP/AGG toward SSIP on TMB addition confirms anion withdrawal from Li^+ shell.
 - B–O stretch of TMB ($\approx 1360 \text{ cm}^{-1}$): shift on FSI^- coordination to boron confirms B–N interaction.
 - Li^+ –solvent stretch ($300\text{--}500 \text{ cm}^{-1}$): should be unchanged, confirming TMB is not entering Li^+ shell.
3. Estimate CIP/AGG fraction from Raman deconvolution. Compute SCE from population fractions. Identify the ratio at which $\text{SCE} \approx 1.466$.
4. Measure CE_{RT} at 25°C and CE_{LT} at -20°C at the target ratio: 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2} , 1.0 mAh cm^{-2} .
5. Apply falsification conditions below.

6.3 Performance Predictions

$$\text{CE}_{\text{RT}}^{\text{pred}} = 73\% \text{ (Regime 1 geometry-driven baseline at SCE*)}$$

$$\text{CE}_{\text{LT}}^{\text{pred}} = 61\% \text{ (LT band at SCE*)}$$

The DOL baseline (81% RT, 58% LT) already exceeds the Regime 1 equation’s RT prediction at its current SCE of 1.41 ($\text{CE}_{\text{RT}}^{\text{eq}} = 74\%$). This suggests DOL has film-forming or SEI contributions beyond the pure solvation geometry effect. The framework prediction is for the solvation geometry contribution; the experimental result may show higher absolute CE values while still confirming the directional prediction.

6.4 Falsification Conditions

The following conditions were committed to the repository at <https://github.com/Eric-Robert-Lawson/attractor-oncology> before any experimental measurement. The commit timestamp is the priority record.

- **Primary falsification:** Framework falsified if CE_{RT} and CE_{LT} do *not both improve* as TMB fraction increases from 0 toward the target ratio. Both must move in the predicted direction simultaneously.
- **Secondary falsification:** Framework falsified if at the ratio confirmed by Raman to have $SCE \approx 1.466$: $CE_{RT} < 87\%$ OR $CE_{LT} < 60\%$.
- **Mechanistic falsification (anion withdrawal):** Mechanism falsified if Raman shows *no* shift in the FSI^- symmetric stretch on TMB addition — indicating TMB is not withdrawing FSI^- from theSSIP/CIP equilibrium.
- **Mechanistic falsification (Li^+ shell integrity):** Mechanism falsified if Raman shows a shift in the Li^+ -solvent stretch region on TMB addition — indicating TMB is entering the Li^+ primary shell and adding rather than removing a coordination class.

Remark 6.1. The four-level falsification structure of Candidate 3 is more granular than Candidate 4 (Preprint 2) because the anion receptor mechanism has two distinct failure modes: TMB does not coordinate FSI^- at all (first mechanistic falsification), or TMB does coordinate FSI^- but also coordinates Li^+ directly, adding rather than removing a coordination class (second mechanistic falsification). Both are experimentally distinguishable by Raman. The four-level structure maps the complete failure mode space of the mechanism.

7 Lewis Acidity Search: TMB Within Accessible Chemical Space

7.1 Search Criteria

The boron ester class was identified in the molecular search (MU Molecular Universe protocol, full record at <https://github.com/Eric-Robert-Lawson/attractor-oncology>, file `MU_Molecular_Search_Complete`) as Region 3: the anion receptor region. The search within this region evaluated candidates against the four selection criteria of Proposition 2.

7.2 TMB as the Primary Candidate

Within the boron ester class, the candidates evaluated were:

Table 2: Boron ester class candidates evaluated as anion receptors. TMB is the only candidate satisfying all four criteria simultaneously.

Molecule	Lewis acidity	Non-coord. Li^+	Electrochem. stable	Physical compat.	Status
TMB $[B(OMe)_3]$	✓	✓	✓	✓	Selected
TEB $[B(OEt)_3]$	✓	✓	✓	Marginal	Alternate
TPB $[B(OPh)_3]$	✓	✓	✓	×	Excluded (solid at RT)
BF_3 etherate	Strong	×	×	×	Excluded (too strong)
$B(C_6F_5)_3$	Very strong	×	×	✓	Excluded (too strong)

TEB (triethyl borate, $[B(OEt)_3]$) satisfies three of four criteria. Its physical compatibility is marginal: boiling point $118^\circ C$ (adequate), but its Lewis acidity is slightly lower than TMB due to the greater electron-donating character of ethyl vs. methyl groups. TEB is named as an alternate candidate: if TMB experiments confirm the mechanism but the TMB boiling point ($68^\circ C$) creates

practical issues in open-cell testing, TEB provides a direct comparison at reduced acidity. The comparison of TMB vs. TEB at equal volume fractions constitutes an internal Lewis acidity gradient test analogous to the Candidate 4b design in Preprint 2.

Stronger Lewis acids (BF_3 etherate, $\text{B}(\text{C}_6\text{F}_5)_3$) are excluded because their Lewis acidity is sufficient to coordinate Li^+ directly, violating criterion (ii) and adding rather than removing a coordination class. The mechanism requires moderate, selective Lewis acidity — strong enough to compete with FSI^- coordination to Li^+ , weak enough not to coordinate Li^+ itself.

7.3 Geometric Closure of the Anion Receptor Search

The boron ester class is the only commercially accessible Lewis acid class satisfying all four criteria simultaneously. The search exhausted the following classes: boron esters, boron halide etherates, triaryl boranes, aluminium alkyls, and silicon Lewis acids. All non-boron-ester classes fail on criterion (ii) (Li^+ coordination) or criterion (iii) (electrochemical stability).

Within the boron ester class, TMB is the primary candidate and TEB the sole alternate. The anion receptor search space is therefore closed: two candidates, one mechanism, two experiments.

8 Structural Asymmetry Between the Anion Receptor and ESP-Matching Mechanisms

The ESP-matching mechanism (introduced in the companion paper, same repository) and the anion receptor mechanism introduced here are both rationally designed approaches to $\text{SCE}^* = 1.466$. They are structurally asymmetric in every dimension that matters for experimental design:

Table 3: Structural asymmetry between the two SCE navigation mechanisms. The two mechanisms are orthogonal: they act on independent sources of SCE and are indicated by opposite diagnostic conditions.

Property	ESP-Matching Mechanism	Anion Receptor Mechanism
Direction of approach	From below SCE^* (shell too ordered)	From above SCE^* or anion-driven excess diversity
Source of SCE addressed	Solvent class diversity — the number of coordination classes present	Anion participation diversity — the CIP/AGG fraction of the population
Mechanism action	Adds a new coordination class (structurally distinct co-solvent)	Removes an existing coordination contributor (sequesters FSI^-)
Effect on p_{max}	Decreases (population splits across classes)	Increases modestly (SSIP fraction rises)
Effect on n_{sig}	Increases	Decreases
Primary reagent	Co-solvent from distinct chemical class (TMP)	Lewis acid anion receptor (TMB)
Raman diagnostic	ESP-matched co-solvent stretch shifts (P=O for TMP, C–O for DME)	FSI^- symmetric stretch shifts (SSIP→CIP→AGG ratio)
Failure mode if wrong direction applied	Raises SCE further above SCE^* in an already disordered system	Lowers SCE further below SCE^* in an already ordered system
Canonical realisation	$\text{LiFSI/DME} + \text{TMP}$	$\text{LiFSI/DOL} + \text{TMB}$

The asymmetry is not merely descriptive. It has a practical consequence: the direction diagnosis

must precede the mechanism selection. Applying the wrong mechanism — adding a new coordination class to a DOL/TMB system that already has adequate diversity, or adding an anion receptor to a DME system that does not have excess anion participation — will move SCE away from SCE* rather than toward it.

The complete rational navigation toolkit for SCE* consists of:

1. Diagnose direction: measure SCE, determine if system is above or below SCE*, identify the primary source of excess or deficit diversity.
2. Select mechanism: if below and solvent-geometry-driven, apply ESP matching. If above or anion-participation-driven, apply anion reception.
3. Apply the mechanism with the canonical reagent.
4. Verify by Raman that the mechanism is operative before measuring CE.
5. Confirm CE response against falsification conditions.

This five-step protocol is the operational consequence of the two mechanisms together. It converts the abstract navigation instruction — “move SCE toward SCE*” — into a concrete, executable, and fully falsifiable experimental design.

The full geometric structure showing how the two mechanisms, all six candidate formulations, the fixed point derivation, and the molecular search records relate to one another is navigable from the repository index at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

9 Discussion

9.1 The Precision of Near-Approach

DOL at SCE = 1.41 is 0.056 SCE units from SCE*. This near-approach has a consequence that distinguishes this candidate from others in the series: the predicted CE changes on reaching SCE* are small in absolute terms. The primary experimental signal is the simultaneous directionality of both CE values, not their magnitude.

This is the correct experimental test for a near-approach system. A large absolute change would suggest the baseline was far from optimal. A small simultaneous improvement is the signature of a system that was already close and needed only fine-tuning — which is exactly the mechanistic claim.

9.2 The DOL Film-Forming Contribution

DOL is known to participate in SEI formation beyond its solvation geometry contribution. DOL ring-opening polymerisation at the Li metal surface produces a flexible, LiF-containing SEI component that is mechanically compliant and ionically conductive. This film-forming contribution is separate from the SCE mechanism and may explain why the DOL baseline CE_{RT} (81%) exceeds the Regime 1 equation prediction at SCE = 1.41 (74%).

The SCE framework does not model film-forming chemistry. It models the solvation geometry contribution to CE. The film-forming contribution is a parallel mechanism that adds to the geometry-driven CE floor rather than replacing it. The experimental prediction — both CEs improve on TMB addition — holds regardless of the film-forming contribution, because the film-forming chemistry is not changed by TMB addition at the target ratio.

9.3 The TEB Comparison as a Built-In Gradient Test

The identification of TEB as the alternate candidate (Section 7) creates a built-in gradient test analogous to Candidate 4b in the companion paper. If TMB at 4:1 DOL:TMB achieves $\text{SCE} \approx 1.466$ and TEB at 4:1 DOL:TEB achieves a lower SCE (because TEB is a weaker Lewis acid and sequesters less FSI^-), the Lewis acidity dependence of the anion receptor mechanism is confirmed. This test requires no additional experimental apparatus: it is a direct comparison of two solutions that differ only in the Lewis acid identity.

9.4 What the Mechanism Does Not Do

The anion receptor mechanism does not:

- Increase ionic conductivity. TMB does not contribute Li^+ transport and sequestering FSI^- may modestly reduce conductivity. Conductivity measurements are therefore a useful parallel characterisation but are not the primary test of the framework.
- Change the solvent geometry. The DOL coordination mode is preserved. If DOL ring-opening SEI chemistry is operative, it continues unchanged.
- Guarantee high absolute CE without salt and additive optimisation. The geometry-driven prediction is a floor, not a ceiling.
- Apply to systems where FSI^- is not participating in the primary shell. In systems with SSIP-dominant coordination (e.g., HCE/LHCE), TMB addition has no anion to withdraw and the mechanism is inoperative. The direction diagnosis must confirm CIP/AGG presence before the anion receptor is applied.

9.5 Limitations

- The population fractions in Table 1 are estimated from the three-configuration simplification. Full MD simulation of the LiFSI/DOL/TMB system is required for quantitative SCE determination at the target ratio.
- The Raman protocol identifies CIP/AGG fractions approximately. Quantitative population analysis requires spectral deconvolution with reference spectra for each coordination state, which must be established separately.
- TMB’s volatility (bp 68 °C) requires careful handling in open-air formulation and may limit practical deployment. TEB (bp 118 °C) is the practical alternate.
- The prediction of no Li^+ coordination by TMB is based on steric and Lewis acidity arguments and existing literature. It should be confirmed directly by the Li^+ –solvent Raman diagnostic specified in the falsification conditions.
- The regime classification of LiFSI/DOL/TMB has not been confirmed experimentally. If CE_{RT} exceeds 90% on TMB addition, the system has crossed into Regime 2 and the Regime 2 equations apply. The protocol must check for this and respond accordingly.

10 Conclusion

We have introduced the **anion receptor mechanism** as the second rationally designed principle for navigating toward the Li metal battery performance fixed point $SCE^* = 1.466$.

The mechanism operates as follows: a Lewis acid additive (trimethyl borate, TMB) selectively coordinates FSI^- anion in solution via a B–N Lewis acid–base interaction, reducing free anion activity and depopulating CIP and AGG configurations in the Li^+ primary solvation shell. The population distribution contracts. SCE moves toward SCE^* . Solvent geometry is preserved.

DOL is the indicated base solvent: its anion-participation-driven population diversity places its SCE close to SCE^* ($\Delta SCE = +0.056$), and the fine-tuning required is within the operating range of the anion receptor mechanism. TMB is the canonical anion receptor: it satisfies all four selection criteria simultaneously, and its boron-ester-class Lewis acidity profile provides the selectivity required for the mechanism to operate without entering the Li^+ shell.

Candidate 3 (LiFSI 1.0 M in DOL/TMB, 4:1 v/v) is pre-registered with four-level falsification conditions that test the mechanism independently of the performance outcome.

The anion receptor mechanism is structurally asymmetric with respect to the ESP-matching mechanism: they act on independent sources of SCE, are indicated by opposite diagnostic conditions, and together constitute the complete rational navigation toolkit for SCE^* .

The full geometric structure of the framework — showing how both mechanisms, all candidate formulations, the fixed point derivation, the molecular search, and the statistical confirmation relate to one another — is navigable from the repository index at

<https://github.com/Eric-Robert-Lawson/attractor-oncology>.

Data and derivation record availability. All pre-registered predictions, falsification conditions, population estimates, Lewis acidity analysis, and derivations referenced in this paper are archived with full timestamped commit history at:

<https://github.com/Eric-Robert-Lawson/attractor-oncology>

The repository commit history constitutes the priority and pre-registration record for all claims in this paper.

Relationship to other papers in this series. This is Preprint 3 of the SCE Framework series. Preprint 1 (foundation, fixed point derivation) and Preprint 2 (ESP-matching mechanism, DME/TMP) are available at the same repository. All papers in the series are self-contained. The geometric structure connecting all papers, mechanisms, candidates, and derivations is navigable from the repository index.

Competing interests. The author declares no competing financial interests.

Author contributions. E.R.L.: conceptualisation, derivation, Lewis acidity analysis, writing.

A TMB Physical and Chemical Properties Summary

B DOL Physical and Chemical Properties Summary

C Raman Diagnostic Protocol

The following Raman peaks are the primary mechanistic diagnostics for Candidate 3. All assignments from literature [12, 2].

Table 4: Physical and chemical properties of TMB relevant to electrolyte formulation.

Property	Value	Source
CAS number	121-43-7	—
Melting point	−29 °C	CRC Handbook [3]
Boiling point	68 °C	CRC Handbook [3]
Density (25 °C)	0.915 g mL ^{−1}	CRC Handbook [3]
Miscibility with DOL	Complete at all ratios	This work
Lewis acidity class	Moderate boron ester	—
Electrochemical stability	Stable 0–4.5 V vs. Li/Li ⁺	[6]
SEI contribution	LiBO ₂ (ionically conductive)	[6]
Commercial availability	Standard reagent	Sigma-Aldrich, TCI

Table 5: Physical and chemical properties of DOL relevant to electrolyte formulation.

Property	Value	Source
CAS number	646-06-0	—
Melting point	−95 °C	CRC Handbook [3]
Boiling point	78 °C	CRC Handbook [3]
Density (25 °C)	1.060 g mL ^{−1}	CRC Handbook [3]
Dielectric constant	7.1	[10]
Donor number	18	[4]
SCE (LiFSI 1M, this series)	1.41	Preprint 1 dataset
SEI contribution	Ring-opening polymerisation, LiF-containing film	[5]
Commercial availability	Standard reagent	Sigma-Aldrich, TCI

Table 6: Raman diagnostic assignments for LiFSI/DOL/TMB mechanistic confirmation.

Peak	Assignment	Expected change	Interpretation
~740 cm ^{−1}	FSI [−] sym. stretch	Shift from CIP/AGG to SSIP on TMB addition	Confirms FSI [−] withdrawal from shell
~1360 cm ^{−1}	TMB B–O stretch	Shift to lower wavenumber on FSI [−] coordination	Confirms B–N interaction open
300–500 cm ^{−1}	Li ⁺ –solvent stretch	No shift on TMB addition	Confirms TMB not entering Li ⁺
~830 cm ^{−1}	DOL C–O–C stretch	No shift on TMB addition	Confirms DOL coordination geometry preserved

D Pre-Registered Falsification Conditions (Verbatim)

The following conditions are reproduced from the repository commit record. The commit timestamp predates this document.

Candidate 3 (LiFSI 1.0M, DOL/TMB 4:1 v/v):

- *Primary*: Framework falsified if CE_{RT} and CE_{LT} do not both improve as TMB fraction increases from 0 toward the target ratio.
- *Secondary*: Framework falsified if at the ratio confirmed by Raman to have SCE ≈ 1.466:

$\text{CE}_{\text{RT}} < 87\%$ OR $\text{CE}_{\text{LT}} < 60\%$.

- *Mechanistic (anion withdrawal)*: Mechanism falsified if Raman shows no shift in the FSI^- symmetric stretch on TMB addition.
- *Mechanistic (Li^+ shell integrity)*: Mechanism falsified if Raman shows a shift in the Li^+ -solvent stretch region on TMB addition.

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